Article Title

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

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1 Introduction

Turots Nusantara constitutes an Islamic intellectual heritage preserved in manuscripts and classical books across the Indonesian archipelago. Many of these manuscripts were written using Arabic script adapted to local languages, such as Pegon (Javanese) and Jawi (Malay). Efforts in digitization and automatic reading (OCR) are important for preservation, text retrieval, and philological studies. However, in old scanned manuscripts, the automation process is often disrupted by unstable image quality, for example uneven page backgrounds (sometimes dark, sometimes bright), ink bleeding, and varying stroke thickness. These conditions make early stages such as binarization and segmentation inconsistent and directly affect the quality of letter-form representation [1, 2].

Progress in Pegon OCR has begun to emerge, for example through the recognition of Pegon characters in typed manuscripts using synthetic datasets and real annotations, as well as through the evaluation of several approaches adapted from Arabic-script OCR. The results show that deep learning-based methods can surpass conventional approaches in certain scenarios [3]. However, in degraded old scanned manuscripts, the main bottleneck often appears before the recognition stage: variations in illumination and background, ink bleed, and changes in stroke thickness cause binarization results to vary across pages and regions, making the letter structures that should be consistent difficult to preserve [4].

A number of previous studies in Arabic script segmentation and recognition indicate that the main problems are still related to the cursive nature of the writing, connected characters, and the sensitivity of methods to image quality. Reviews of modern Arabic OCR emphasize that Arabic-based scripts have high structural complexity, especially due to inter-letter connectivity, variations in letter position, and the presence of diacritics [5–7]. At the segmentation level, approaches such as seam carving are able to handle touching characters, but they are highly sensitive to noise and image degradation [8]. Hybrid approaches based on morphology and connected-component analysis have also shown good results under certain conditions, but their performance is still affected by handwriting complexity and input quality [9].

In the context of degraded Pegon/Turots documents, a structure-based representation is needed so that character modeling does not depend on raw pixel intensity. One widely used representation is skeletonization, namely reducing binary strokes into a centerline to preserve the core shape along with its topological aspects. Modern skeleton literature emphasizes that the skeleton is a compact representation useful for shape description and segmentation, but its quality is sensitive to noise and binarization artifacts, which in practice often cause disconnections, path deviations from the stroke center, and unstable centerline structures [10–12]. This problem becomes more critical in Pegon writing because its forms are cursive, strokes can become thicker or thinner, and small components such as dots or harakat are easily attached or lost.

Under these conditions, conventional skeletonization tends to produce spurs on noisy edges and unstable centerline paths. To stabilize the skeleton path so that it remains consistently on the "ridge" of stroke thickness, the Distance Transform (DT) provides a local radius signal, namely the distance to the nearest boundary, which can be used to guide the centerline to follow the DT ridge. Approaches that utilize distance transformation or edge information as reinforcement for skeleton extraction have also been reported to be effective in improving skeleton quality, especially when object shapes are complex and noisy [13]. Thus, this kind of geometric information is relevant for strengthening structural representation in degraded Pegon manuscript images.

In addition to the general problem of noise, Pegon manuscripts exhibit very specific shape failures in looping letters, for example a case that often appears in the form of wawu—when the loop cavity closes due to thickened ink or binarization results that close the hole. In a "blob"-like circular condition without a hole, skeletonization often forms a chord line across the middle of the blob or a branching structure that no longer represents the loop, thereby biasing the letter shape and potentially disrupting segmentation and recognition. To detect this case systematically, this study uses circularity as a detector of "how round" a Pegon letter component is [14]. Highly circular components can be treated as candidates for closed loops so that the skeleton path can be redirected to more closely resemble a representative circular trajectory.

On the other hand, strengthening shape structure from the beginning is also important because the historical manuscript domain often faces limitations in adequate annotated data. The release of open-access datasets for ancient Arabic writing confirms that validated historical data, in the form of image-text ground truth pairs, are still relatively scarce, even though they are greatly needed for the training, evaluation, and generalization of deep learning-based models [15, 16]. Therefore, a structure-based approach that can improve representation quality without dependence on large amounts of data becomes a practical strategy for the digitization of Pegon/Turots manuscripts [17, 18]. Therefore, a shape-representation strategy that can improve the quality of character structure without relying on large datasets is important for the digitization of Pegon/Turots manuscripts.

Table 1 Comparison of Related Studies and This Research

Ref	Task / Focus	Method	Limitation	Relevance to This Research
[19]	Layout segmentation of ancient Arabic manuscripts	One-shot learning for document layout segmentation using only one labeled page per manuscript	Focuses on the page/layout level and has not yet addressed character structure or loop recovery	Relevant as a pre-OCR stage, but this study goes further to the character-component level in degraded Pegon manuscripts
[20]	Arabic handwritten document OCR pipeline	Differentiable Binarization + Adaptive Scale Fusion + CNN-BiLSTM-CTC	More oriented toward the recognition pipeline; does not explicitly handle closed-loop topology or unstable local skeletons	This study focuses on structural representation before OCR, particularly skeleton stabilization and loop shape recovery
[21]	Historical handwritten Arabic text recognition	Transformer-based encoder-decoder HTR that uses spatial attention for cursive characters and diacritics	Strong in recognition, but still data/model-heavy and not designed to improve character shape structure at the image level	This study offers a lighter and more interpretable approach for small and noisy Pegon data
[22]	Arabic handwritten dataset building for archival documents	Construction of the REJD dataset for Arabic archival documents, including OCR data preparation	The main contribution lies in dataset provision, not in improving the structure of degraded letter components	Supports the argument that data are important, but this study targets a different problem: character structure recovery under limited-data conditions
[23]	Identification of paleographic curvature using skeletonization and key point detection	A skeleton-based method for segmenting connected Jawi handwritten characters using centroid, intersection, and loop detection	The main contribution lies in applying lightweight structural analysis for Jawi character segmentation in small manuscript datasets	This study supports the importance of structural character analysis in limited-data manuscript processing, but it focuses on segmentation rather than stroke reconstruction or degraded component recovery.
[24]	Application of the traveling salesman problem to optimize skeletonization and stroke reconstruction	A skeleton-based Pegon manuscript method that combines Zhang-Suen skeletonization with a modified Greedy TSP to reconstruct stroke paths	The main contribution lies in reconstructing handwriting trajectories by allowing controlled revisits on branching and looping skeleton structures	This study supports the importance of structural path optimization in Arabic-derived manuscript analysis, but it focuses on stroke reconstruction rather than detailed character component recovery.
This Study	Optimization of degraded Pegon character structure	Pre-processing, CCA, component classification, Zhang-Suen thinning, Distance Transform-based skeleton refinement, circularity-based circular blob detection, loop development, and skeletonization	This study emphasizes improving character structure at the component level, so validation is still limited to the quality of visual representation and not yet to text recognition accuracy	Proposes a lightweight and interpretable structural approach to stabilize skeletons, preserve loop topology, and produce more appropriate character representations for degraded Pegon manuscripts

Therefore, a lightweight, interpretable, and effective structural representation strategy is needed for small and noisy Pegon manuscript data. In this study, a skeleton structure optimization approach through Zhang-Suen thinning and circularity analysis is proposed to represent characters more accurately in degraded Pegon manuscript images. Zhang-Suen thinning is used to obtain a single-pixel skeleton while maintaining stroke connectivity, while circularity is used to identify highly rounded components as closed-loop candidates that lose holes due to image degradation or binarization processes. This approach is expected to provide a more suitable character structure representation to support the analysis of historical Pegon manuscripts.

2 Methods

In this study, a skeleton structure optimization method is proposed for degraded Pegon manuscript images by integrating Zhang-Suen thinning to obtain the basic structure of character strokes, Distance Transform (DT) to support the stability of the geometric representation of strokes, circularity analysis to identify circular components that have lost their holes, as well as connected component analysis and bounding boxes to define the spatial boundaries of each analyzed component. The proposed method is organized into several main stages, namely pre-processing, component extraction, baseline and stroke radius estimation, Zhang-Suen- and DT-based skeletonization, circular blob detection, loop skeleton development, and skeleton evaluation. The overall process flow in this study is presented in Figure 1.

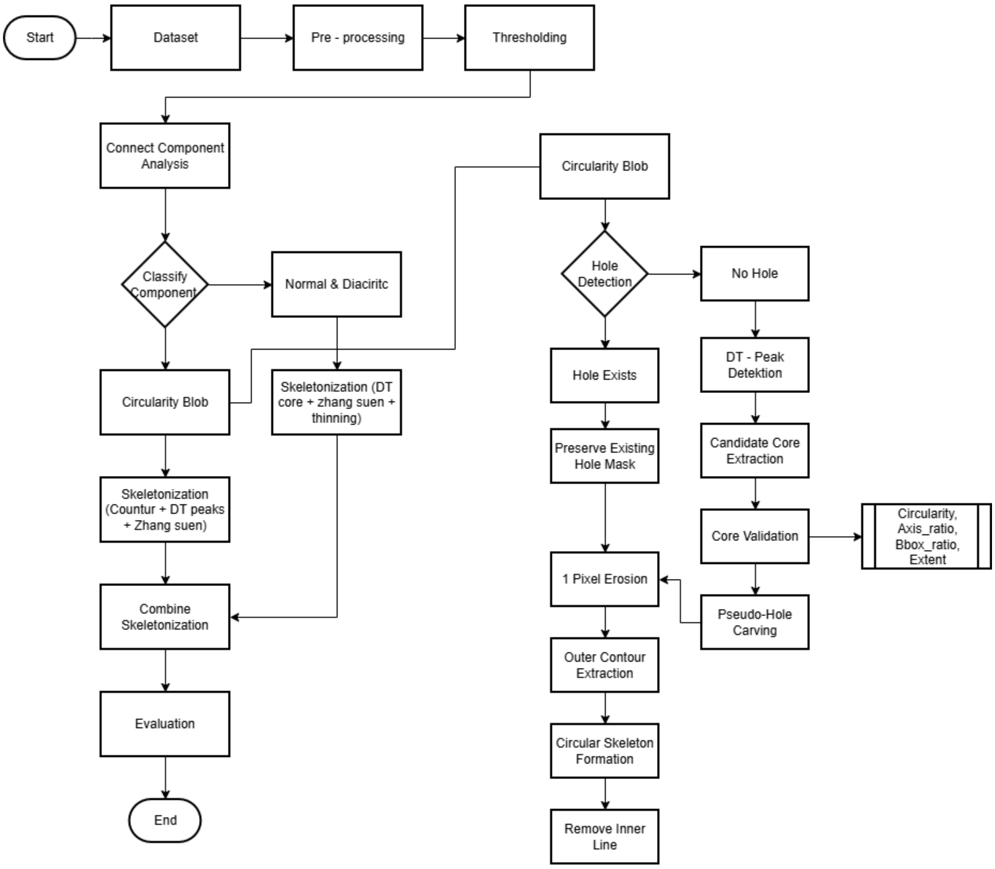


Fig. 1 Research flowchart.

2.1 Dataset

The dataset used in this study consists of scanned images of the Syair Perang Johor manuscript written in Jawi script [25]. The digitization process was carried out by the Research Center for Artificial Intelligence and Cyber Security, National Research and Innovation Agency (BRIN). Overall, the manuscript consists of page and line images in PNG format. Ground truth annotation was conducted independently by four annotators who were capable of reading Jawi manuscripts. Differences in the annotation results were then resolved through consensus-based discussions facilitated by experts in Jawi manuscript studies, who also performed the final validation. This procedure was designed to ensure the reliability of the annotations as well as their consistency with the scholarly principles of Jawi manuscript studies.

2.2 Pre-Processing

Pre-processing was carried out to clarify the separation between the writing foreground and the image background. At this stage, the image was enlarged by a factor of 2, then inverted, converted to grayscale, and its contrast was enhanced using CLAHE. After that, thresholding was performed using a combination of Otsu threshold, adaptive Gaussian threshold, and morphological operations so that the resulting binary image became more stable for component analysis. The binarization process can be expressed as follows.

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases} \tag{1}$$

where $f(x,y)$ is the grayscale pixel intensity and T is the threshold value.In this formulation, pixels satisfying the threshold condition are represented as foreground, while the remaining pixels are treated as background. This binary decision model follows the general principle of threshold-based document binarization[26].

2.3 Connected Component Analysis (CCA)

After binarization, the foreground image is analyzed using Connected Component Analysis (CCA) to group mutually connected pixels into text components. In document image analysis, connected components are commonly used to detect and extract text parts or objects that are spatially separated. In literature relevant to handwritten text segmentation, connected components are used together with morphological operations and bounding boxes to detect text units. In addition, in binary image-based region analysis, component labeling is also commonly performed using 8-way pixel connectivity.

The set of foreground pixels in a binary image is defined as

$$F = \{p = (x,y) \in \Omega \mid B(p) = 1\} \tag{2}$$

where $B(p)$ is the pixel value in the binary image resulting from pre-processing at location $p = (x,y)$, with $B(p) = 1$ denoting foreground and $B(p) = 0$ denoting background. Meanwhile, Ω denotes the image domain, namely the set of all pixel coordinates in the image [27].

$$N_8(p) = \{q = (x',y') \in \Omega \mid |x - x'| \leq 1,\ |y - y'| \leq 1,\ q \neq p\} \tag{3}$$

This definition states that all pixels located around p, whether horizontally, vertically, or diagonally, are included in the 8-neighborhood of p. The use of 8-neighborhood connectivity is suitable for cursive and handwritten scripts because stroke connections may occur not only horizontally and vertically, but also diagonally [28].

A connected component C_k is then defined as the set of foreground pixels that can be reached from a seed S_k through an 8-neighborhood path, namely

$$C_k = \{q \in F \mid \exists (p_0, p_1, \ldots, p_n),\ p_0 = S_k,\ p_n = q,\ p_i \in F,\ p_{i+1} \in N_8(p_i),\ \forall i = 0, \ldots, n-1\} \tag{4}$$

With this definition, one component C_k is a maximal region in which all pixels are mutually connected through an 8-neighborhood path. This notation is more commonly used to explain connected components than writing a brief logical relation without an explicit definition. The component area is calculated as the number of foreground pixels in the component, namely

$$A = \sum_{(x,y)\in C_k} 1 \tag{5}$$

while the component centroid is computed from image moments as

$$C_x = \frac{m_{10}}{m_{00}}, \quad C_y = \frac{m_{01}}{m_{00}} \tag{6}$$

where m_{00} states the area of the component [29].

2.4 Component Classification

Component classification is performed to determine the component type before entering the skeletonization stage. In this study, each component is classified as *diacritic*, *circular_blob*, or *normal*. The decision rules are determined based on area, distance to the baseline (dy), circularity ($circ$), axis ratio ($axis$), and bounding box ratio ($bbox\_ratio$). The baseline is obtained from the median vertical position of body components, while the global stroke size is computed from the median value of the foreground distance transform and is denoted as r_{stroke}.

2.4.1 Component Classification Features

1. The first feature is area, namely the number of foreground pixels in a component, which is expressed as

   $$A = \sum_{(x,y)\in C_k} 1 \tag{7}$$

   The area value is used to distinguish small components such as diacritics from larger components.
2. The second feature is the distance to the baseline, namely the vertical distance of the component centroid to the baseline, which is computed as

   $$dy = |c_y - baseline_y| \tag{8}$$

   where c_y is the vertical coordinate of the component centroid and $baseline_y$ is the baseline position. Components with large dy values tend to be located far from the main body of the character.
3. The third feature is circularity, namely a measure of component roundness based on the area and contour perimeter, expressed as

   $$circ = \frac{4\pi A}{p^2} \tag{9}$$

where A is the component area and p is the contour perimeter. A *circ* value approaching 1 indicates a shape that is increasingly round.Circularity is used because roundness is an important shape descriptor for detecting compact or circle-like objects, and it is relevant for identifying blob-like components that may correspond to closed-loop structures [30].

4. The fourth feature is axis ratio, which is computed from the ratio between the minor axis and major axis obtained from contour fitting, namely

$$axis = \frac{b}{a}, \qquad b \leq a \tag{10}$$

An *axis* value approaching 1 indicates that the component shape is more symmetric and closer to a circle.

5. The fifth feature is bounding box ratio, namely the ratio of the short side to the long side of the component bounding box, expressed as

$$bbox\_ratio = \frac{\min(w, h)}{\max(w, h)} \tag{11}$$

where w is the bounding box width and h is its height. A larger *bbox_ratio* value indicates a more compact component shape. The combination of circularity, axis ratio, and bounding box ratio is used to describe circular blob candidates from complementary geometric perspectives. Circularity measures roundness, axis ratio captures elongation or symmetry, and bounding box ratio reflects spatial compactness. Using more than one shape descriptor is useful because degraded manuscript components may have noisy or imperfect contours, making a single descriptor less reliable for component classification [31].

To define the area threshold, the following formulation is used:

$$A_{\text{threshold}} = \pi r_{\text{stroke}}^2 \tag{12}$$

Then, the maximum area limits for diacritics and blobs are defined as follows:

$$A_{\text{diac,max}} = \frac{\pi}{4}\, r_{\text{stroke}}^2 \tag{13}$$

$$A_{\text{blob,max}} = 20\pi r_{\text{stroke}}^2 \tag{14}$$

Meanwhile, the distance threshold from the baseline is expressed as

$$dy_{\max} = \max\left(2.5\, r_{\text{stroke}},\ 0.35\, H_{\text{body}}\right) \tag{15}$$

where H_{body} is the median height of the body text. The use of r_{stroke} is motivated by distance transform analysis, where each foreground pixel is assigned a distance value to the nearest background or boundary pixel. Therefore, distance transform values can be used as local thickness or stroke-radius information for adapting component thresholds. The constants $\frac{1}{4}$, 20, 2.5, and 0.35 are empirical parameters in this study. These constants are not treated as universal formulas from previous studies, while the underlying idea of adapting thresholds to stroke thickness is supported by distance-transform-based geometric analysis [32].

2.4.2 Component Classification Rules

A component is classified as *diacritic* if it has a small area and is located sufficiently far from the baseline, namely

$$A < A_{\mathrm{diac,max}} \quad \text{and} \quad dy > dy_{\max} \tag{16}$$

A component is classified as *circular_blob* if the component area is still within the blob limit and its shape is sufficiently round. The roundness level is computed from the average of three shape features, namely circularity, axis ratio, and bounding box ratio, expressed as

$$score_{\mathrm{round}} = \frac{score_{\mathrm{circ}} + score_{\mathrm{axis}} + score_{\mathrm{bbox}}}{3} \tag{17}$$

A component is considered sufficiently round if

$$score_{\mathrm{round}} \geq 0.50 \tag{18}$$

and its position is not too far from the baseline, namely

$$dy \leq 2dy_{\max} \tag{19}$$

Thus, a component is classified as *circular_blob* if it satisfies

$$A \leq A_{\mathrm{blob,max}} \;\wedge\; score_{\mathrm{round}} \geq 0.50 \;\wedge\; dy \leq 2dy_{\max} \tag{20}$$

Components that do not satisfy either the *diacritic* or the *circular_blob* criteria are classified as *normal*. In brief, the component decision function can be written as

$$label(C_k) = f(x) = \begin{cases} diacritic, & A < A_{\mathrm{diac,max}} \;\wedge\; dy > dy_{\max} \\ circular\_blob, & A \leq A_{\mathrm{blob,max}} \;\wedge\; score_{\mathrm{round}} \geq 0.50 \;\wedge\; dy \leq 2dy_{\max} \\ normal, & other \end{cases} \tag{21}$$

The above formula defines an interpretable rule-based approach because the use of area, centroid position, height, width, and bounding box ratio follows the connected-component-based and shape-based analysis, while threshold values are determined empirically according to the observed characteristics of the Pegon/Jawi manuscript components [14, 30].

2.5 Skeletonization

Skeletonization is performed to reduce the stroke thickness into a one-pixel representation while preserving topological connectivity. The basic method used is Zhang-Suen thinning. In this algorithm, a pixel p is removed if it satisfies

$$2 \leq B(p) \leq 6 \tag{22}$$

$$A(p) = 1 \tag{23}$$

where $B(p)$ is the number of active neighbors in 8-connectivity and $A(p)$ is the number of $0 \rightarrow 1$ transitions in the circular ordering of the neighbors. In the first sub-iteration, the following conditions apply:

$$p_2 p_4 p_6 = 0, \qquad p_4 p_6 p_8 = 0 \tag{24}$$

Meanwhile, in the second sub-iteration, the following conditions apply:

$$p_2 p_4 p_8 = 0, \qquad p_2 p_6 p_8 = 0 \tag{25}$$

The iteration is continued until no more pixels are removed. In addition to the pure Zhang-Suen skeleton, the system also forms a comparison skeleton based on the ridge of the distance transform, and the result is then used in a hybrid strategy. Skeletonization is appropriate in this context because it converts binary objects into compact centerline representations that support structural analysis, while Zhang-Suen thinning is used to produce single-pixel skeletons while preserving connectivity[33].

2.5.1 Skeletonization for Circular Blob

Components of the *circular_blob* type are processed specially because the target skeleton is a loop shape with an empty center. If the component does not yet have a hole, the system first forms a pseudo-hole from the local distance transform peak. The core region is formed by

$$core = \{(x, y) \mid DT(x, y) \geq \alpha r_0\} \tag{26}$$

where $DT(x, y)$ is the distance transform value, r_0 is the local radius at the peak, and α is the core threshold constant. This formulation uses high distance-transform values to identify the central or thick region of the component, because distance transform represents the distance from a foreground point to the nearest boundary or background point. Therefore, the local peak of the distance transform can be used as an indicator of the medial region of a thick or closed component [14, 32]. After that, the foreground is eroded by 1 pixel, and the skeleton is formed from the outer contour of the component. In simplified form, the circular blob stage can be written as

$$F_e = Erode(F) \tag{27}$$

$$S_{\text{circ}} = \partial F_e \tag{28}$$

where ∂F_e denotes the outer contour of the object. The lines in the inner part of the circle are then removed so that the center of the blob remains empty.The use of the outer contour is relevant because contour-based extraction is commonly used to represent object boundaries after component separation in binary region analysis. This special treatment is used to restore loop-like structures in circular components so that the resulting skeleton does not form an incorrect chord line across the center of the blob[13, 32].

2.5.2 Skeletonization for Normal and Diacritic Components

Normal and diacritic-type components are processed using a Hybrid DT-core + Zhang-Suen strategy. The original foreground is first thinned using Zhang-Suen to obtain the basic skeleton

$$S_{\text{orig}} = ZS(F) \tag{29}$$

Then, the local core-based skeleton is obtained from the foreground that has undergone pseudo-hole carving

$$S_{\text{core}} = ZS(F^c) \cap core \tag{30}$$

where F^c is the foreground after carving and C is the mask of the preserved core area. The final skeleton of a normal component is formed by combining the original skeleton and the core skeleton, and then thinning it again

$$S_{\text{normal}} = ZS(S_{\text{orig}} \cup S_{\text{core}}) \tag{31}$$

This approach keeps the skeleton of normal components thin while still being able to preserve the core structure of the component. Zhang-Suen-based thinning is relevant because thinning algorithms are used to produce single-pixel skeletons while preserving object connectivity[33].

2.5.3 Loop Development and Skeleton Cleaning

After the main skeleton is obtained, non-*circular_blob* components may still go through a distance transform-based loop development stage. Thick areas are deduced from skeleton pixels located at positions where

$$DT(x, y) > T_{\text{loop}} \tag{32}$$

in the implementation, the following value is used

$$T_{\text{loop}} = 5.0 \tag{33}$$

In that area, the original thresholded shape is expanded again into a local loop, and then thinned using Zhang-Suen. After that, final cleaning is performed by removing skeleton lines located inside closed loops, while still preserving the outer loop lines [32, 34].

2.6 Combine Skeletonization

After the skeleton of each component is obtained, all skeletons are merged into a single global skeleton. This process is carried out using a binary union operation on all component skeletons. For *circular_blob*, the inner part of the blob is kept empty so that skeletons from other components do not enter the central loop area.

In general, the skeleton merging can be written as

$$S_{\text{final}} = \bigvee_{k=1}^{n} S_k \tag{34}$$

where S_k is the skeleton of the k-th component. The result of this merging becomes the final skeleton of the image.

2.7 Evaluation

The evaluation stage is carried out to measure how well the skeleton optimization method produces a correct representation of Pegon character structure. The evaluation is performed by comparing the skeleton produced automatically by the system with ground truth that has been defined manually. This process uses several standard evaluation metrics in the field of pattern recognition, namely accuracy, precision, recall, and F1-score.

3 Results and Discussion

3.1 Pre-Processing

Figure 2 shows the pre-processing stages, which include grayscale conversion, thresholding, and clean thresholding. These stages effectively improve the separation between the writing foreground and the background, thereby producing a more stable input for structural analysis in the subsequent stage. The grayscale image preserves the distribution of the main strokes while reducing image complexity, the thresholding result emphasizes the shape of the character body by separating the text from the background, while the clean thresholding result further suppresses unwanted noise so that the main character components become easier to distinguish. However, some very small foreground details may still potentially weaken during the cleaning stage, especially in areas affected by degradation or uneven ink distribution. This condition indicates that the pre-processing stage needs to balance noise reduction and detail preservation so that the structural characteristics of the writing remain preserved for the subsequent component analysis and skeletonization stages.

(a) (b) (c)

Fig. 2 Pre-processing stage: (a) grayscale image, (b) thresholding image, and (c) cleaned thresholding image.

3.2 Circularity

The results of the circularity stage are shown in Figure 3, which illustrates the difference between the input binary image and the result of circular component identification. At this stage, the binary image serves as the initial input, which is then analyzed to recognize components that have round or nearly round shape characteristics. Thus, circularity is used to select character parts that potentially form loops and require special structural treatment.

(a) (b)

Fig. 3 Circularity stage: (a) binary input image, (b) circularity result image.

In Figure 3(a), the input binary image can be seen before the circularity analysis is performed. At this stage, the character strokes are still represented as the foreground in the image, so the closed loop parts have not yet appeared as holes in the middle of the component. In other words, the area that should form a circular structure is still merged as a binary mass, so the hole information in the inner part of the loop has not yet appeared explicitly.

In Figure 3(b), after the circularity analysis is applied, components that tend to have a circular shape become more clearly identified. This result shows that circularity is able to recognize loops that were previously closed in the binary image, so that the circular structure can be restored with a hole in its center. With the formation of that hole, the component representation becomes more consistent with the original geometry of the character, especially in parts that indeed have a loop shape.

3.3 Ablation

Table 2 presents the sensitivity test results of component classification under different threshold scales for two approaches, namely the proposed method (*This work*) and the comparison method (*Fixed threshold*). In this table, the scale does not refer to image size or pixel resolution, but to the level of looseness or strictness of the threshold used in the component classification process. Smaller scale values represent looser thresholds, whereas larger scale values represent stricter thresholds.

Table 2 Comparison of scalable thresholding and fixed thresholding.

Method	Scale	Diacritic	Circular Blob	Normal	Stability
This work	0.7	0	13	13	1.000
	1.0	0	13	13	1.000
	1.4	0	13	13	1.000
Fixed threshold	0.7	0	18	8	0.808
	1.0	0	13	13	1.000
	1.4	0	11	15	0.923

The *Diacritic*, *Circular Blob*, and *Normal* columns show the number of components assigned to each class. *Diacritic* denotes small components located relatively far from the main body of the text. *Circular Blob* denotes components with round or nearly round shape characteristics that therefore require special treatment during the skeletonization stage. Meanwhile, *Normal* denotes components that do not satisfy the criteria of the previous two classes and are treated as the main text components. The *Stability* column indicates the consistency of classification results when the threshold scale changes. A stability value closer to 1 indicates that the component labels remain consistent, whereas a lower value indicates that some components shift from one class to another when the parameter changes.

The results in Table 2 show that the proposed method has very high stability. Across all scale variations, namely 0.7, 1.0, and 1.4, the numbers of components classified as *Diacritic*, *Circular Blob*, and *Normal* remain unchanged at 0, 13, and 13, respectively, with a stability value of 1.000. This finding indicates that the proposed method is able to preserve its classification decisions consistently, even when the threshold becomes either looser or stricter. Therefore, the proposed method can be considered more robust to changes in classification parameters.

In contrast, the fixed-threshold approach shows higher sensitivity to scale changes. At the scale of 1.0, its classification result is identical to that of the proposed method. However, when the scale is reduced to 0.7, the number of *Circular Blob* components increases while the number of *Normal* components decreases, accompanied by a drop in stability to 0.808. This indicates that a looser threshold causes more components to be classified as circular blobs. Conversely, when the scale is increased to 1.4, the number of *Circular Blob* components decreases and the number of *Normal* components increases, with a stability value of 0.923. This suggests that a stricter threshold causes some components previously identified as circular blobs to shift into the normal category.

Overall, these results confirm that the proposed method is more stable than the fixed-threshold approach in maintaining consistent component classification across different threshold scales. This indicates that the proposed method is more adaptive to parameter variations and is therefore more reliable for degraded Pegon manuscript images, which often exhibit substantial variation in component shape and stroke thickness. The Diacritic, Circular Blob, and Normal columns show the number of components assigned to each class. Diacritic represents small components whose positions are relatively separated from the main body of the writing. Circular Blob represents components that have round or nearly round shape characteristics, so they are treated specifically in the skeletonization stage. Meanwhile, Normal represents components that do not meet the characteristics of the previous two classes and are treated as the main components of the writing. The Stability column indicates the level of classification stability when the threshold scale is changed. A stability value close to 1 indicates that the component labels remain consistent, while a lower value indicates that some components change class when the parameters are modified.

The results in Table 2 show that the proposed method has a very high level of stability. Across all scale variations, namely 0.7, 1.0, and 1.4, the number of components classified as Diacritic, Circular Blob, and Normal remains the same, namely 0, 13, and 13, with a Stability value of 1.000. This finding indicates that the proposed method is able to maintain classification decisions consistently even when the threshold is made looser or stricter. Thus, the proposed method can be considered more robust to changes in classification parameters.

In contrast, the Fixed Threshold approach shows higher sensitivity to scale changes. At a scale of 1.0, the classification results are still the same as those of the proposed method. However, when the scale is reduced to 0.7, the number of Circular Blob components increases, while the number of Normal components decreases, accompanied by a decrease in the Stability value to 0.808. This indicates that a looser threshold causes more components to tend to fall into the circular blob category. Conversely, when the scale is increased to 1.4, the number of Circular Blob components decreases and the number of Normal components increases, with a Stability value of 0.923. This condition indicates that a stricter threshold causes some components that were previously recognized as circular blobs to shift into the normal category.

Overall, this table confirms that the proposed method is more stable than the fixed threshold in maintaining the consistency of component classification across various threshold scales. These results indicate that the proposed method is more adaptive to changes in classification parameters, making it more reliable for use on Pegon manuscript images that have variations in component shapes and stroke thickness.

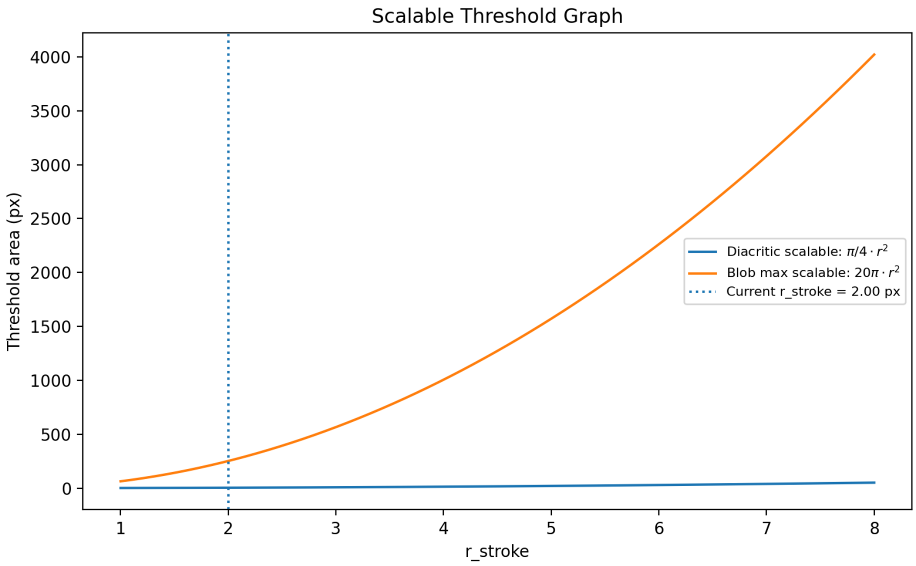


Fig. 4 Scalable threshold result graph

At this stage, a comparison graph between scalable threshold and fixed threshold is also presented, where r_{stroke} represents the stroke thickness size of the character, namely the measure that describes the width of the main strokes in the binary image. Meanwhile, the area threshold refers to the foreground component area limit used in the component classification process. The area in question is calculated based on the number of foreground pixels in each connected component, so the area threshold in this study represents the component area limit, not the bounding box area or the size of an entire letter as a whole.

Figure 4 shows the behavior of the area threshold, which is scalable with respect to changes in the value of r_{stroke}. As r_{stroke} increases, the area threshold for the diacritic and circular blob categories also increases. This indicates that the classification boundary is adjusted proportionally to the stroke thickness. Thus, when characters have thicker strokes, the system automatically enlarges the component area limit that can still be classified as diacritic or circular blob. This characteristic shows that the scalable approach is adaptive to variations in writing morphology.

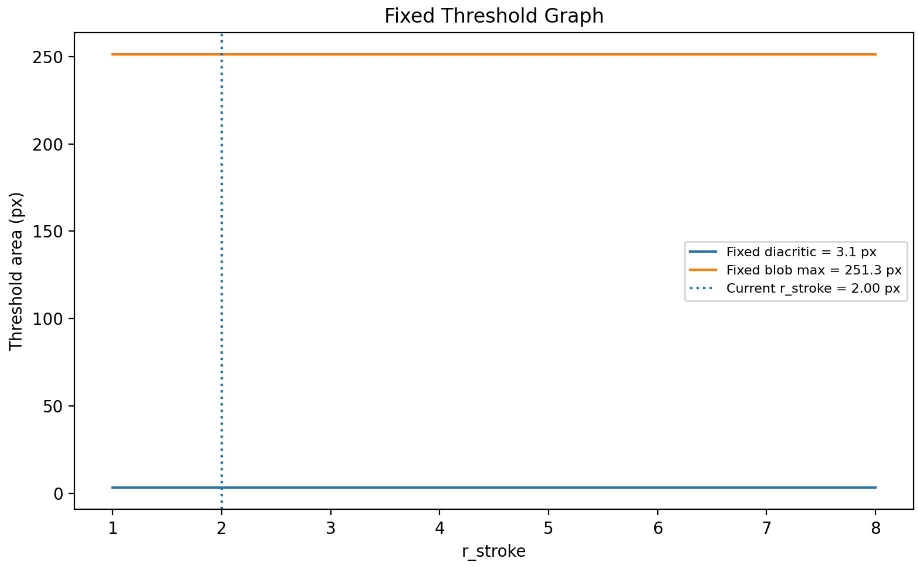


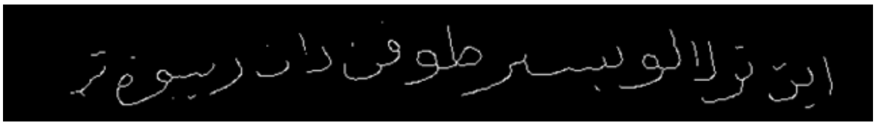
Fig. 5 Behavior of fixed threshold area with respect to r_{stroke}. Unlike the scalable approach, the threshold remains constant across the entire range of r_{stroke}.

Figure 5 shows the behavior of the fixed threshold area with respect to changes in r_{stroke}. In contrast to the scalable approach, the threshold in this figure remains constant over the entire range of r_{stroke}. This indicates that the classification boundary does not change even when the stroke thickness varies. As a consequence, the fixed-threshold approach is only suitable under a particular reference condition and becomes less proportional when the stroke is either thinner or thicker than the reference condition.

3.4 Skeletonization

The skeletonization results are shown in Figure 6, which present a comparison between the skeleton without circularity and the skeleton with circularity. In general, both results successfully reduce stroke thickness into a thinner representation, but the quality of the resulting structure shows a clear difference, especially in the formation of loop parts and the similarity of the shape to the original character.

(a)



(b)

Fig. 6 Skeletonization results: (a) skeleton without circularity, (b) skeleton with circularity.

In Figure 6(a), the skeleton produced without circularity has not yet been able to fully represent the circular parts. Some components that visually should form loops still appear open or not completely closed, so the resulting skeleton structure does not yet fully follow the original shape of the character. As a result, the representation produced still tends to lose shape characteristics in the circular parts, especially in components that experience hole closure due to image degradation.

In contrast, in Figure 6(b), after circularity is applied, the skeleton structure shows results that are closer to the original shape of the character. Parts that previously did not form loops now appear clearer and more complete, so the skeleton not only becomes thinner, but also more representative of the geometry of the source character. This more consistent loop formation shows that circularity plays an important role in recognizing circular components, and then guiding the skeleton to follow a path that is more consistent with the original structure of the writing.

The difference in the results of the two subfigures shows that the addition of circularity makes a significant contribution to the quality of the skeleton, especially for characters or components that have loop shapes. Without circularity, skeletonization tends to produce a representation that is not yet able to adequately restore circular structures. Meanwhile, with circularity, loops can be formed better so that the final skeleton shape becomes more stable, more informative, and more similar to the original character image. Thus, the use of circularity in the skeletonization stage has been proven to improve the conformity of the structural representation to the shape of degraded Pegon characters.

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